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RAG-Driven Customer Support Assistant System.
Deliverable 1- HLD

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1. System Overview

1.1 Problem Definition:

- Traditional QA systems and chatbots often generate **hallucinated or incorrect answers** because they rely only on pre-trained knowledge.
- This project solves the problem by implementing a **Retrieval-Augmented Generation (RAG)** system that:
    - Extracts knowledge from PDFs
    - Retrieves relevant context
    - Generates **accurate, context-based answers**

1.2 Scope of the System:

- Upload and process PDF documents
- Convert text into embeddings
- Store in vector database (ChromaDB)
- Accept user queries
- Retrieve relevant chunks
- Generate contextual answers using LLM
- Provide fallback using Human-in-the-Loop (HITL)

1.3 Core Objective :

- Improve **accuracy of responses**
- Reduce hallucination
- Enable **intelligent document-based QA system**
- Provide scalable AI architecture

2. System Architecture Diagram

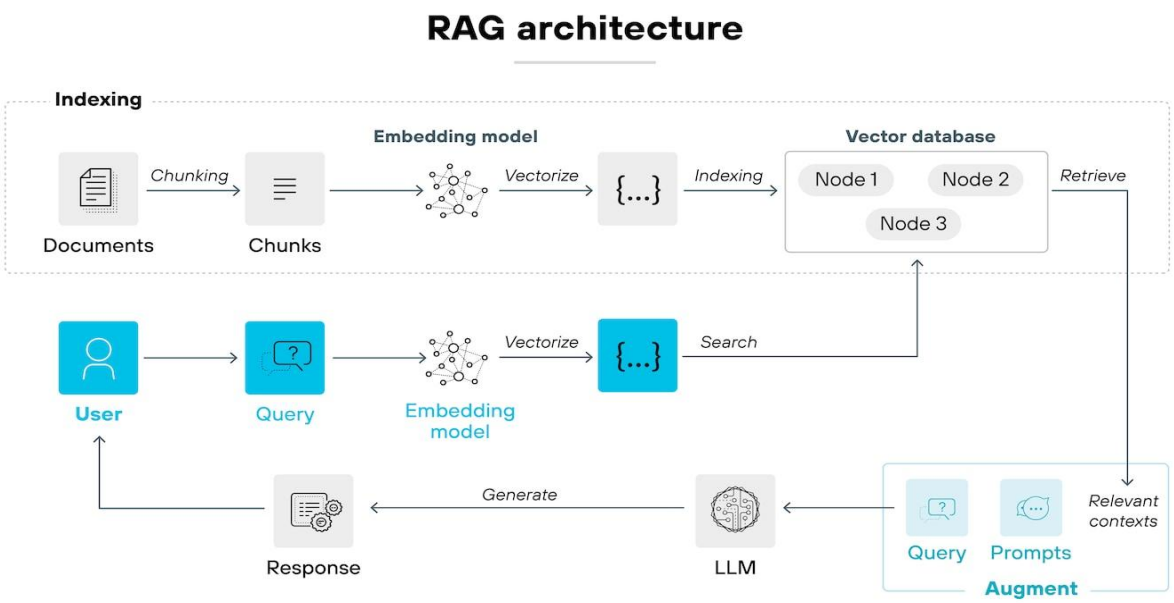

Figure 1: Dual-Phase RAG Architecture & LangGraph Workflow Logic

2.1 The Data Ingestion & Transformation Tier

This phase represents the **offline processing pipeline**, where documents are converted into structured intelligence.

- **Source Extraction:** PDF documents are processed using document loaders to extract raw text while preserving structure.
- **Semantic Chunking:** Text is split into smaller overlapping chunks using recursive splitting techniques to maintain context continuity.
- **Vectorization Engine:** Each chunk is converted into vector embeddings using embedding models to capture semantic meaning.

2.2 The Knowledge Retrieval & Vector Storage Tier

This layer manages storage and retrieval of document knowledge.

- **ChromaDB Vector Store:** Stores embeddings and performs efficient similarity search.
- **Contextual Retriever:** Retrieves Top-K relevant chunks based on user query similarity.

2.3 The Logic Orchestration & Routing Tier

This layer controls system flow and decision-making.

- **LangGraph Workflow Engine:** Maintains stateful execution and manages query lifecycle.
- **Intent Routing:** Determines whether query should be answered automatically or escalated.
- **HITL Integration:** Handles uncertain or complex queries with human validation.

3. Detailed Component Decomposition

3.1 The Logic Engine (LangGraph):

- Manages state transitions across workflow nodes.
- Controls execution flow based on conditions and query state.

3.2 Knowledge Retrieval Module:

- Performs semantic similarity search.
- Fetches and ranks relevant document chunks.

3.3 LLM Synthesis Layer:

- Generates responses using retrieved context.
- Ensures output is grounded and meaningful.

3.4 HITL Escalation Module:

- Handles low-confidence queries.
- Allows human review and correction.

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4. System Interconnectivity & Data Lifecycle

4.1 Automatic Ingestion Lifecycle:

Document → Extraction → Chunking → Embedding → Storage

4.2 Real-Time Execution Cycle:

User Query → Retrieval → Context → LLM → Response

4.3 Feedback & Escalation Workflow:

Low Confidence → HITL → Verified Answer → Output

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5. Architectural Rationalization

5.1 Selection of ChromaDB

- Efficient vector storage and retrieval
- Lightweight and developer-friendly
- Fast similarity search

5.2 Implementation of LangGraph:

- Enables stateful workflows
- Supports branching logic
- Allows human intervention

5.3 LLM Selection :

- Provides natural language understanding
- Generates context-aware responses

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6. Optimization & Scalability Strategy

6.1 Large-Scale Document Handling:

- Efficient chunking and indexing
- Metadata filtering

6.2 Query Load Handling:

- Optimized retrieval (Top-K tuning)
- Caching frequently asked queries

6.2 Latency Optimization:

- Balanced chunk size
- Efficient embedding models
- Fast vector search

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7. Conclusion

The proposed RAG-based system provides a **robust, scalable, and intelligent solution** for document-based question answering.

By integrating:

- Retrieval (ChromaDB)
- Workflow orchestration (LangGraph)
- LLM reasoning
- HITL validation

the system ensures **high accuracy, reliability, and enhanced user experience**.